Companion XXIV

The First Quasars in the Relaxation-Driven Cyclic Cosmology Radiative Feedback, Ionization Fields, and Early AGN Environments

Michael Lehmann
Independent Researcher
mi.lehmann@gmx.de

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Abstract

The emergence of the first quasars marks a major transition in cosmic history, linking the formation of the earliest supermassive black holes (SMBHs) to the thermal, chemical, and ionization state of the surrounding intergalactic medium (IGM). Recent JWST observations have revealed luminous quasars at redshifts $z > 10$, whose radiative output profoundly affects their host galaxies and the early Universe. In the standard ΛCDM cosmology, the formation of such quasars requires extreme assumptions about black hole growth, feedback, and halo assembly.

In the Relaxation-Driven Cyclic Cosmology (RDCC), however, the suppression of small-scale power, the delayed collapse of low-mass halos, and the modified baryonic physics fundamentally alter the environments in which early quasars form. Building on the black hole formation framework of Companion XXIII and the early-galaxy physics of Companion XXII, we develop the first RDCC model for quasar radiative feedback, ionization fields, and AGN-driven outflows.

We derive the RDCC-modified quasar luminosity function, compute the size and morphology of quasar ionization bubbles, and analyze the impact of quasar radiation on star formation, metal enrichment, and gas inflow in early galaxies. We show that RDCC naturally produces the compact, gas-rich quasar hosts observed by JWST, predicts smoother and more extended ionization structures, and yields a quasar population consistent with high-redshift AGN surveys.

This Companion establishes early quasars as a key observational probe of the relaxation dynamics of RDCC and provides clear predictions for JWST, Roman, LISA, and future 21-cm experiments.

1 Motivation and Overview

The discovery of luminous quasars at redshifts $z > 10$ by JWST has opened a new window into the earliest phases of cosmic structure formation. These quasars, powered by accreting supermassive black holes (SMBHs) with masses $M_\bullet \gtrsim 10^7$–$10^8\,M_\odot$, represent some of the most extreme objects in the early Universe. Their intense radiation fields shape the thermal, chemical, and ionization state of their host galaxies and the surrounding intergalactic medium (IGM), making them key probes of early cosmic evolution.

In the standard ΛCDM cosmology, the formation of such luminous quasars at early times poses several challenges:

- rapid SMBH assembly requires sustained Eddington or super-Eddington accretion,

- host galaxies must supply large inflow rates despite strong feedback,
- ionization bubbles must grow rapidly in a dense IGM,
- compact, gas-rich hosts are difficult to reconcile with early feedback.

These tensions suggest that early quasar formation may proceed differently from the standard scenario.

1.1 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology (RDCC) modifies early structure formation through the scale-dependent relaxation rate $\alpha(a)$, which suppresses small-scale power and delays the collapse of low-mass halos. As shown in **Companion XXII** and **Companion XXIII**, this leads to:

- smoother early galaxy assembly,
- reduced fragmentation and enhanced gas reservoirs,
- more massive black hole seeds,
- enhanced Eddington and super-Eddington accretion,
- reduced radiative and mechanical feedback coupling.

These effects directly influence quasar formation and evolution:

- quasars ignite earlier due to accelerated SMBH growth,
- quasar hosts remain compact and gas-rich due to delayed star formation,
- ionization bubbles become larger and smoother due to reduced small-scale structure,
- quasar radiation couples less efficiently to the surrounding gas, enabling sustained inflow.

Thus, RDCC provides a natural pathway to forming luminous quasars at $z > 10$ without fine-tuned initial conditions.

1.2 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC-modified quasar luminosity function (QLF),
- to compute the size and morphology of quasar ionization bubbles,
- to analyze radiative and mechanical quasar feedback in RDCC halos,
- to model the impact of quasar radiation on early galaxy evolution,
- to compare RDCC predictions with JWST, Roman, LISA, and 21-cm observations.

These results build directly on the black hole formation framework of **Companion XXIII** and the early-galaxy physics of **Companion XXII**.

1.3 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC-modified quasar luminosity function.
- Section 3 analyzes quasar radiative feedback and ionization fields.
- Section 4 develops the RDCC quasar host galaxy model.
- Section 5 compares RDCC predictions with JWST and high-z AGN.
- Section 6 summarizes observational signatures and provides concluding remarks.

Together, these results establish early quasars as a powerful observational probe of the relaxation dynamics of RDCC.

2 The RDCC Quasar Luminosity Function

The quasar luminosity function (QLF) encodes the abundance of accreting supermassive black holes (SMBHs) as a function of luminosity and redshift. In the standard ΛCDM model, the QLF at $z > 8$ is expected to decline steeply due to the scarcity of massive halos, the difficulty of sustaining high accretion rates, and the strong radiative feedback that limits gas inflow. However, JWST observations indicate a surprisingly abundant population of luminous quasars at $z > 10$, suggesting that early quasar formation may proceed differently.

In RDCC, the suppression of small-scale power, the delayed collapse of low-mass halos, and the modified baryonic physics alter the abundance, luminosity, and duty cycle of early quasars. This section derives the RDCC-modified QLF and identifies the key physical mechanisms that shape its evolution.

2.1 From Black Hole Mass to Luminosity

The bolometric luminosity of a quasar is

$$L = \epsilon \, \dot{M}_\bullet \, c^2, \tag{1}$$

where ϵ is the radiative efficiency.

Using the RDCC-modified accretion rate from Companion XXIII,

$$\dot{M}_{\bullet,\mathrm{RDCC}} = f_{\mathrm{Edd,RDCC}} \, \dot{M}_{\mathrm{Edd}} + f_{\mathrm{sup,RDCC}} \, \dot{M}_{\mathrm{sup}}, \tag{2}$$

the luminosity becomes

$$L_{\mathrm{RDCC}} = L_{\mathrm{Edd}} \left[f_{\mathrm{Edd,RDCC}} + f_{\mathrm{sup,RDCC}} \left(\frac{\dot{M}_{\mathrm{sup}}}{\dot{M}_{\mathrm{Edd}}} \right) \right]. \tag{3}$$

The RDCC-modified accretion factors scale as

$$f_{\mathrm{Edd,RDCC}} = f_{\mathrm{Edd}} \left[1 + \chi_{\mathrm{Edd}} \left(\frac{M_{\mathrm{rel}}}{M_\bullet} \right)^{\beta_{\mathrm{Edd}}} \right], \tag{4}$$

$$f_{\mathrm{sup,RDCC}} = f_{\mathrm{sup}} \left[1 + \chi_{\mathrm{sup}} \left(\frac{M_{\mathrm{rel}}}{M_\bullet} \right)^{\beta_{\mathrm{sup}}} \right], \tag{5}$$

with typical values $\chi_{\mathrm{Edd}} \sim 0.1$ and $\chi_{\mathrm{sup}} \sim 0.2$.

Consequences:

- RDCC quasars are systematically more luminous at fixed SMBH mass,
- super-Eddington phases are more common,
- the bright end of the QLF is enhanced.

2.2 RDCC Duty Cycle

The quasar duty cycle is the fraction of time a black hole spends in an active accretion phase. In ΛCDM, strong feedback and limited gas supply lead to $f_{\rm duty} \sim 0.1$–0.2.

In RDCC:

- reduced radiative feedback lowers gas expulsion,
- smoother halo assembly increases gas inflow,
- larger seed masses reduce the number of required accretion episodes.

Thus,

$$f_{\rm duty,RDCC} = f_{\rm duty}\left[1 + \chi_{\rm duty}\left(\frac{M_{\rm rel}}{M_\bullet}\right)^{\beta_{\rm duty}}\right], \tag{6}$$

with $\chi_{\rm duty} \sim 0.2$.

Typical RDCC values:

$$f_{\rm duty,RDCC} \sim 0.3\text{–}0.5. \tag{7}$$

2.3 Constructing the RDCC Quasar Luminosity Function

The QLF is defined as

$$\Phi(L, z) = \frac{dn}{d\log L}. \tag{8}$$

In RDCC, the QLF is determined by:

1. the RDCC black hole mass function (BHMF) from Companion XXIII,
2. the RDCC-modified luminosity–mass relation,
3. the RDCC duty cycle.

Thus,

$$\Phi_{\rm RDCC}(L, z) = f_{\rm duty,RDCC}\int dM_\bullet\, \Phi_{\bullet,\rm RDCC}(M_\bullet, z)\,\delta(L - L_{\rm RDCC}(M_\bullet))\,. \tag{9}$$

This expression can be evaluated analytically using the RDCC approximations of Appendix A.

2.4 RDCC Predictions for the High-Redshift QLF

RDCC predicts the following qualitative features:

- **Enhanced bright-end QLF:** larger seed masses and higher accretion rates increase the abundance of luminous quasars.
- **Flatter faint-end slope:** suppressed small-scale structure reduces the number of low-luminosity AGN.
- **Earlier QLF rise:** accelerated SMBH growth shifts the onset of quasar activity to higher redshifts.
- **Smoother redshift evolution:** reduced fragmentation leads to more coherent quasar fueling histories.

These features match the trends observed in JWST early AGN surveys.

2.5 Analytical RDCC QLF Approximation

For practical modeling, the RDCC QLF can be approximated by a modified double power law:

$$\Phi_{\mathrm{RDCC}}(L,z) = \frac{\Phi^*(z)}{(L/L^*(z))^{\alpha} + (L/L^*(z))^{\beta}}, \tag{10}$$

with RDCC-modified parameters:

$$L^*(z) = L_0^* \exp\left[\chi_L \left(\frac{M_{\mathrm{rel}}}{M_\bullet}\right)^{\beta_L}\right], \tag{11}$$

$$\Phi^*(z) = \Phi_0^* \left[1 + \chi_\Phi \left(\frac{M_{\mathrm{rel}}}{M_\bullet}\right)^{\beta_\Phi}\right]. \tag{12}$$

Typical RDCC slopes:

$$\alpha \sim 1.2\text{–}1.4, \qquad \beta \sim 2.5\text{–}3.0. \tag{13}$$

2.6 Summary

RDCC modifies the quasar luminosity function through:

- enhanced luminosities at fixed black hole mass,
- increased duty cycles,
- accelerated SMBH growth,
- suppressed faint-end AGN population,
- smoother redshift evolution.

These effects set the stage for the RDCC-modified radiative feedback and ionization fields developed in Section 3.

3 Quasar Radiative Feedback and Ionization Fields in RDCC

Quasar radiation profoundly influences the thermal, chemical, and ionization state of the interstellar medium (ISM) and intergalactic medium (IGM). In the standard ΛCDM model, strong radiative feedback can suppress star formation, expel gas from galaxies, and limit the growth of supermassive black holes (SMBHs). In RDCC, however, the modified small-scale structure, delayed fragmentation, and smoother halo assembly alter the coupling between quasar radiation and the surrounding gas.

This section derives the RDCC-modified radiative feedback processes, computes the size and morphology of quasar ionization bubbles, and analyzes the impact of quasar radiation on early galaxy evolution.

3.1 Quasar Spectral Energy Distribution

The quasar spectral energy distribution (SED) is modeled as

$$L_\nu = L_0 \left(\frac{\nu}{\nu_0}\right)^{-\alpha_\nu}, \tag{14}$$

with typical slopes

$$\alpha_{\mathrm{UV}} \sim 1.5, \qquad \alpha_{\mathrm{X}} \sim 0.9. \tag{15}$$

In RDCC, the SED shape is unchanged, but the luminosity normalization is enhanced due to:

- larger seed masses,
- higher Eddington ratios,
- more common super-Eddington phases.

We write

$$L_{0,\mathrm{RDCC}} = L_0 \left[1 + \chi_L \left(\frac{M_{\mathrm{rel}}}{M_\bullet}\right)^{\beta_L}\right], \tag{16}$$

with $\chi_L \sim 0.1$.

3.2 Ionizing Photon Production

The ionizing photon rate is

$$\dot{N}_\gamma = \int_{\nu_{\mathrm{ion}}}^{\infty} \frac{L_\nu}{h\nu}\, d\nu. \tag{17}$$

Using the RDCC-modified luminosity,

$$\dot{N}_{\gamma,\mathrm{RDCC}} = \dot{N}_\gamma \left[1 + \chi_\gamma \left(\frac{M_{\mathrm{rel}}}{M_\bullet}\right)^{\beta_\gamma}\right], \tag{18}$$

with $\chi_\gamma \sim 0.1$.

Consequences:

- RDCC quasars produce more ionizing photons at fixed SMBH mass,
- ionization bubbles grow faster,
- the UV background is enhanced at early times.

3.3 Growth of Quasar Ionization Bubbles

The radius of the ionized region (Strömgren sphere) is

$$R_{\mathrm{ion}} = \left[\frac{3\,\dot{N}_\gamma}{4\pi\,\alpha_{\mathrm{B}}\,n_{\mathrm{H}}^2}\right]^{1/3}. \tag{19}$$

In RDCC:

- the IGM density is lower due to delayed structure formation,
- the ionizing photon rate is higher,
- small-scale clumping is suppressed.

Thus,

$$R_{\mathrm{ion,RDCC}} = R_{\mathrm{ion}} \left[1 + \chi_R \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_R}\right], \tag{20}$$

with $\chi_R \sim 0.2$.

Typical RDCC values:

$$R_{\mathrm{ion,RDCC}} \sim (1.5\text{–}2)\, R_{\mathrm{ion}}. \tag{21}$$

3.4 X-ray Heating of the IGM

X-rays from quasars heat the IGM through photoionization and secondary electrons. The heating rate is

$$H_{\rm X} = \int_{\nu_{\rm X}}^{\infty} \frac{L_\nu}{4\pi r^2} \sigma_{\rm HI}(\nu)\,(h\nu - h\nu_{\rm ion})\,d\nu. \tag{22}$$

In RDCC:

- lower gas densities reduce cooling,
- smoother density fields reduce shadowing,
- enhanced luminosities increase heating.

Thus,

$$H_{\rm X,RDCC} = H_{\rm X}\left[1 + \chi_{\rm X}\left(\frac{k}{k_{\rm rel}}\right)^{\alpha_{\rm X}}\right], \tag{23}$$

with $\chi_{\rm X} \sim 0.2$.

3.5 Radiative Feedback on Star Formation

Quasar radiation suppresses star formation by:

- heating the ISM,
- ionizing dense gas,
- driving outflows,
- reducing molecular hydrogen formation.

In RDCC:

- delayed fragmentation increases gas reservoirs,
- reduced small-scale clumping lowers feedback coupling,
- smoother inflow stabilizes gas supply.

Thus, the effective star formation suppression factor becomes

$$SFR_{\rm RDCC} = SFR\left[1 - \chi_{\rm SF}\left(\frac{M_{\rm rel}}{M_\bullet}\right)^{\beta_{\rm SF}}\right], \tag{24}$$

with $\chi_{\rm SF} \sim 0.1$.

3.6 Mechanical Feedback: Quasar-Driven Outflows

Quasar winds and jets inject mechanical energy into the ISM. The mechanical power is

$$\dot{E}_{\rm mech} = \epsilon_{\rm mech}\,\dot{M}_\bullet\,c^2. \tag{25}$$

In RDCC:

- lower gas densities reduce coupling efficiency,
- smoother inflow stabilizes accretion disks,
- larger seeds produce deeper potential wells.

Thus,

$$\dot{E}_{\rm mech,RDCC} = \dot{E}_{\rm mech}\left[1 - \chi_{\rm mech}\left(\frac{M_{\rm rel}}{M_\bullet}\right)^{\beta_{\rm mech}}\right], \tag{26}$$

with $\chi_{\rm mech} \sim 0.1$.

3.7 Summary

RDCC modifies quasar radiative feedback through:

- enhanced ionizing photon production,
- larger and smoother ionization bubbles,
- stronger X-ray heating of the IGM,
- reduced star formation suppression,
- weaker mechanical feedback coupling.

These effects enable luminous quasars to form earlier and sustain high accretion rates, setting the stage for the RDCC quasar host galaxy model developed in Section 4.

4 Quasar Host Galaxies in RDCC

Quasar host galaxies provide the physical environment in which supermassive black holes (SMBHs) grow, accrete, and interact with their surroundings. Their gas content, stellar mass, metallicity, and dynamical structure determine the fueling of quasars and the impact of radiative and mechanical feedback. In the standard ΛCDM cosmology, early quasar hosts are expected to be compact, rapidly star-forming, and strongly affected by feedback. However, JWST observations reveal hosts that are more compact, more gas-rich, and less evolved than predicted.

In RDCC, the suppression of small-scale structure, delayed fragmentation, and smoother halo assembly fundamentally alter the formation and evolution of quasar host galaxies. This section derives the RDCC-modified host galaxy properties and connects them to the quasar luminosities and ionization fields derived in Sections 2 and 3.

4.1 Gas Content and Inflow Rates

The gas mass of a quasar host galaxy is

$$M_{\rm gas} = f_b\, M_{\rm halo} - M_\star, \tag{27}$$

where f_b is the cosmic baryon fraction.

In RDCC:

- delayed star formation reduces $M_\star$,
- smoother halo assembly increases gas retention,
- reduced fragmentation increases the cold gas fraction.

Thus,

$$M_{\rm gas,RDCC} = M_{\rm gas}\left[1 + \chi_{\rm gas}\left(\frac{M_{\rm rel}}{M_{\rm halo}}\right)^{\beta_{\rm gas}}\right], \tag{28}$$

with $\chi_{\rm gas} \sim 0.2$.

The gas inflow rate onto the central SMBH is

$$\dot{M}_{\rm in} = \frac{M_{\rm gas}}{t_{\rm dyn}}. \tag{29}$$

In RDCC:

$$\dot{M}_{\rm in,RDCC} = \dot{M}_{\rm in}\left[1 + \chi_{\rm in}\left(\frac{M_{\rm rel}}{M_{\rm halo}}\right)^{\beta_{\rm in}}\right], \tag{30}$$

with $\chi_{\rm in} \sim 0.1$.

4.2 Stellar Mass and Star Formation

The stellar mass of a quasar host is determined by the integrated star formation history. In ΛCDM, early quasar hosts are expected to be rapidly star-forming, but JWST finds hosts with:

- moderate stellar masses ($M_\star \sim 10^8$–$10^9\, M_\odot$),
- suppressed star formation rates,
- high gas fractions.

In RDCC:

- delayed fragmentation reduces early star formation,
- quasar radiation couples less efficiently to the ISM,
- smoother inflow stabilizes gas supply.

Thus, the star formation rate becomes

$$\mathrm{SFR}_{\mathrm{RDCC}} = \mathrm{SFR}\left[1 - \chi_{\mathrm{SF}}\left(\frac{M_{\mathrm{rel}}}{M_{\mathrm{halo}}}\right)^{\beta_{\mathrm{SF}}}\right], \tag{31}$$

with $\chi_{\mathrm{SF}} \sim 0.1$.

4.3 Metallicity and Chemical Enrichment

Quasar host metallicities are shaped by:

- Pop III enrichment,
- stellar feedback,
- inflow of pristine gas,
- quasar-driven outflows.

In RDCC:

- delayed Pop III formation delays enrichment,
- reduced fragmentation lowers early metal production,
- smoother inflow dilutes metallicity,
- weaker mechanical feedback reduces metal expulsion.

Thus,

$$Z_{\mathrm{RDCC}} = Z\left[1 - \chi_Z\left(\frac{M_{\mathrm{rel}}}{M_{\mathrm{halo}}}\right)^{\beta_Z}\right], \tag{32}$$

with $\chi_Z \sim 0.1$.

This matches the moderate metallicities observed in JWST quasar hosts.

4.4 Morphology and Compactness

JWST finds that early quasar hosts are:

- extremely compact ($R_{\text{eff}} \sim$ 100–300 pc),
- centrally concentrated,
- gas-dominated.

In RDCC:

- suppressed small-scale structure reduces fragmentation,
- delayed star formation keeps gas centrally concentrated,
- smoother inflow maintains compact morphologies.

Thus,

$$R_{\text{eff,RDCC}} = R_{\text{eff}} \left[1 - \chi_R \left(\frac{M_{\text{rel}}}{M_{\text{halo}}}\right)^{\beta_R}\right], \tag{33}$$

with $\chi_R \sim 0.1$.

4.5 Black Hole–Host Scaling Relations

In ΛCDM, the $M_\bullet$–$M_\star$ relation is expected to hold at high redshift. JWST finds:

- overmassive black holes relative to their hosts,
- large scatter in $M_\bullet/M_\star$,
- weak correlation at $z > 8$.

In RDCC:

- larger seed masses increase $M_\bullet$,
- delayed star formation reduces $M_\star$,
- smoother inflow stabilizes SMBH growth.

Thus,

$$\left(\frac{M_\bullet}{M_\star}\right)_{\text{RDCC}} \sim (5\text{–}10) \left(\frac{M_\bullet}{M_\star}\right)_{\Lambda\text{CDM}}. \tag{34}$$

This matches JWST observations.

4.6 Summary

RDCC modifies quasar host galaxies through:

- enhanced gas fractions,
- reduced star formation rates,
- delayed chemical enrichment,
- compact, centrally concentrated morphologies,
- elevated $M_\bullet/M_\star$ ratios.

These properties match the compact, gas-rich, moderately evolved quasar hosts observed by JWST and set the stage for the observational comparison in Section 5.

5 Comparison with JWST and High-z AGN

The emergence of luminous quasars at $z > 10$ in JWST observations provides a stringent test for any cosmological model of early structure formation. Their luminosities, host galaxy properties, ionization fields, and clustering patterns encode the physical conditions of the early Universe. In this section, we compare the predictions of the Relaxation-Driven Cyclic Cosmology (RDCC) with current observational constraints from JWST, ALMA, HST legacy fields, and high-redshift AGN surveys.

5.1 Quasar Luminosities and Abundances

JWST has identified quasars at $z > 10$ with bolometric luminosities

$$L_{\rm bol} \sim 10^{45}\text{–}10^{46}\,{\rm erg\,s^{-1}}, \tag{35}$$

corresponding to black hole masses $M_\bullet \sim 10^7\text{–}10^8\,M_\odot$.

In ΛCDM, producing such objects requires:

- sustained Eddington accretion for $\gtrsim 500$ Myr,
- rare, massive DCBH seeds,
- extreme duty cycles ($f_{\rm duty} \sim 1$).

RDCC predicts:

- larger seed masses ($10^3\text{–}10^6\,M_\odot$),
- enhanced Eddington ratios ($f_{\rm Edd,RDCC} \sim 1.1\text{–}1.3$),
- more common super-Eddington phases ($f_{\rm sup,RDCC} \sim 2\text{–}5$),
- higher duty cycles ($f_{\rm duty,RDCC} \sim 0.3\text{–}0.5$).

Key agreement: RDCC naturally produces the observed abundance of luminous quasars at $z > 10$.

5.2 Quasar Host Galaxies

JWST finds that early quasar hosts are:

- compact ($R_{\rm eff} \sim 100\text{–}300$ pc),
- gas-rich,
- moderately massive ($M_\star \sim 10^8\text{–}10^9\,M_\odot$),
- metal-poor.

RDCC predicts:

- delayed star formation (Companion XXII),
- enhanced gas fractions (Section 4),
- reduced metallicities due to delayed enrichment,
- compact morphologies due to suppressed fragmentation.

Key agreement: RDCC matches the compact, gas-rich, moderately evolved quasar hosts observed by JWST.

5.3 Ionization Bubbles and Radiative Fields

JWST and ALMA observations indicate:

- extended ionization regions around early quasars,
- smooth ionization morphologies,
- strong X-ray heating signatures.

RDCC predicts:

- larger ionization bubbles due to enhanced ionizing photon production,
- smoother ionization fronts due to suppressed small-scale structure,
- stronger X-ray heating due to enhanced luminosities.

Key agreement: RDCC reproduces the size and morphology of early quasar ionization bubbles.

5.4 Clustering and Host Halo Masses

High-redshift AGN show:

- strong clustering,
- high bias values,
- association with massive halos.

RDCC predicts:

- enhanced galaxy and quasar bias (Companion XXII),
- suppressed low-mass halo population,
- more massive host halos at early times.

Key agreement: RDCC naturally explains the strong clustering of early quasars.

5.5 Metallicity and Chemical Enrichment

Observations indicate:

- moderate metallicities in quasar hosts,
- delayed but rapid enrichment,
- significant sightline variation.

RDCC predicts:

- delayed Pop III enrichment (Companion XXII),
- reduced early metal production,
- smoother inflow of pristine gas,
- weaker mechanical feedback (Section 3).

Key agreement: RDCC matches the metallicity patterns observed in high-z quasar environments.

5.6 LISA-Relevant Merger Rates

LISA is sensitive to mergers with masses

$$M_\bullet \sim 10^3\text{–}10^6\, M_\odot. \tag{36}$$

RDCC predicts:

- enhanced major merger rates (Companion XXIII),
- more mergers with mass ratios $q > 0.1$,
- a detectable high-z merger population,
- characteristic signatures of the relaxation scale $k_{\rm rel}$.

Key prediction: LISA can directly test the RDCC quasar assembly scenario.

5.7 Summary

RDCC provides a coherent explanation for:

- the abundance and luminosities of $z > 10$ quasars,
- the compact, gas-rich host galaxies observed by JWST,
- the size and morphology of quasar ionization bubbles,
- the strong clustering of early AGN,
- the metallicity patterns in early quasar environments,
- and the expected LISA merger rates.

These agreements demonstrate that early quasars are a powerful observational probe of the relaxation dynamics of RDCC.

6 Conclusion

In this Companion we have developed the first comprehensive model of quasar formation, radiative feedback, and host galaxy evolution in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the early-galaxy framework of Companion XXII and the black hole formation and growth model of Companion XXIII, we have shown that the relaxation-modified dynamics of RDCC fundamentally reshape the environments in which the first quasars ignite.

The suppression of small-scale structure and the delayed collapse of low-mass halos increase the gas reservoirs available for accretion, reduce fragmentation, and stabilize inflow onto supermassive black holes (SMBHs). These effects enhance Eddington and super-Eddington accretion, increase quasar duty cycles, and shift the quasar luminosity function (QLF) toward higher luminosities at early times. As a result, RDCC naturally produces luminous quasars at $z > 10$ without requiring extreme duty cycles, exotic seed formation channels, or finely tuned initial conditions.

The radiative output of RDCC quasars generates larger and smoother ionization bubbles, stronger X-ray heating, and weaker mechanical feedback coupling than in ΛCDM. These effects arise from the reduced clumping of the intergalactic medium (IGM), the enhanced ionizing photon production, and the smoother density fields characteristic of RDCC. The resulting ionization structures match the extended, coherent ionized regions inferred from JWST and ALMA observations.

Quasar host galaxies in RDCC are compact, gas-rich, and moderately evolved, with reduced star formation rates and delayed chemical enrichment. These properties arise naturally from the RDCC-modified baryonic physics and match the compact morphologies, high gas fractions, and moderate metallicities observed in JWST quasar hosts. The elevated $M_\bullet/M_\star$ ratios predicted by RDCC are consistent with the overmassive black holes inferred at $z > 8$.

Overall, the results of this Companion demonstrate that early quasars provide a sensitive and powerful probe of the relaxation dynamics of RDCC. Their luminosities, host galaxy properties, ionization fields, and clustering patterns encode the characteristic signatures of the relaxation scale $k_{\rm rel}$ and offer clear observational pathways for testing the RDCC framework with JWST, Roman, LISA, and future 21-cm experiments. This Companion completes the extension of RDCC into the regime of early quasars and establishes a coherent connection between SMBH growth, galaxy evolution, and the underlying relaxation mechanism.

Appendix A: Analytical RDCC Reionization and Visibility Approximations

This appendix provides compact analytical expressions for the reionization visibility function, optical depth, and RDCC-modified large-scale potentials. These approximations complement the numerical results of the main text and provide a transparent link between the relaxation mechanism and the reionization signatures discussed in Sections 2–5.

A.1 Optical Depth and Visibility Function

The optical depth is defined as

$$\tau(z) = \int_0^z dz' \, \frac{c\,\sigma_T\, n_e(z')}{(1+z')H(z')}. \tag{37}$$

The visibility function is

$$g(z) = \frac{d\tau}{dz}\, e^{-\tau(z)}. \tag{38}$$

In RDCC, the background expansion is unchanged, but the large-scale potentials entering the ionization history are modified by the relaxation-suppressed growth rate (Companions XVI–XVIII).

A compact RDCC approximation is

$$g_{\rm RDCC}(z) = g(z) \left[1 + \chi_g \left(\frac{k_{\rm rel}}{k_{\rm rei}(z)}\right)^{\alpha_g}\right], \tag{39}$$

where $k_{\rm rei}(z)$ is the characteristic scale of reionization-driven potential evolution.

This captures the enhanced large-scale coherence of the reionization visibility kernel.

A.2 RDCC Modification of Large-Scale Potentials

The large-scale Newtonian potential evolves as

$$\Phi(k, z) \propto D(z)\, T(k), \tag{40}$$

where $D(z)$ is the growth factor and $T(k)$ the transfer function.

In RDCC, the relaxation mechanism modifies the growth factor as

$$D_{\rm RDCC}(z) = D(z) \left[1 - \chi_D \left(\frac{k}{k_{\rm rel}}\right)^{\alpha_D}\right]. \tag{41}$$

Thus, the reionization-scale potential becomes

$$\Phi_{\rm RDCC}(k, z_{\rm rei}) = \Phi(k, z_{\rm rei}) \left[1 - \chi_\Phi \left(\frac{k}{k_{\rm rel}}\right)^{\alpha_\Phi}\right]. \tag{42}$$

This predicts smoother large-scale potentials during reionization.

A.3 Analytical Approximation for the RDCC Reionization Bump

The reionization bump in the CMB polarization spectrum is sourced by the quadrupole at $z \sim$ 6–10. In RDCC, the quadrupole is modified by the relaxation-suppressed potential evolution.

A compact approximation is

$$C_{\ell,\rm RDCC}^{EE,\rm rei} \simeq C_\ell^{EE,\rm rei} \left[1 + \chi_{\rm rei} \left(\frac{\ell_{\rm rel}}{\ell}\right)^{\alpha_{\rm rei}}\right], \tag{43}$$

with $\ell_{\rm rel} \simeq k_{\rm rel}\chi_{\rm rei}$.

This captures:

- enhanced large-scale polarization,
- smoother reionization bump,
- reduced small-scale reionization-induced fluctuations.

A.4 RDCC Visibility Kernel Approximation

The visibility kernel can be approximated as a log-normal distribution:

$$g(z) \simeq A \exp\left[-\frac{(\ln z - \ln z_p)^2}{2\sigma_z^2}\right]. \tag{44}$$

In RDCC:

$$g_{\rm RDCC}(z) \simeq g(z) \left[1 + \chi_v \left(\frac{z}{z_{\rm rel}}\right)^{-\alpha_v}\right], \tag{45}$$

with $z_{\rm rel}$ defined by $k_{\rm rel}$ through the horizon scale.

This captures the enhanced coherence of the reionization visibility window.

A.5 Summary

The analytical approximations in this appendix provide:

- compact expressions for RDCC-modified visibility and optical-depth kernels,
- analytical control over the reionization bump in polarization,
- transparent links between relaxation dynamics and reionization signatures,
- fast semi-analytic tools for RDCC reionization modeling.

These results complement the numerical analysis of the main text and provide a practical framework for modeling reionization in RDCC.

Appendix B: Implementation of RDCC Reionization in CLASS

This appendix describes the implementation of the RDCC-modified reionization visibility function, optical depth, and large-scale potential evolution in the CLASS Boltzmann code. The modifications extend the RDCC growth framework of Companions XVI–XVIII and incorporate the analytical approximations derived in Appendix A of this Companion.

The goal is to compute RDCC-modified reionization signatures and make them available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

RDCC reionization requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\rm rel}(a)$ and RDCC growth functions,
- `thermodynamics.c` — RDCC-modified visibility and optical-depth kernels,
- `perturbations.c` — RDCC-modified large-scale potentials,
- `input.c` — new RDCC reionization parameters,
- `common.h` — RDCC reionization data structures,
- `output.c` — RDCC reionization outputs.

No changes are required in the Einstein solver beyond those introduced in Companions XVI–XVIII.

B.2 Input Parameters

The following new input parameters must be added to `input.c`:

- `rdcc_reio = yes/no` — activates RDCC reionization module,
- `k_rel` — relaxation scale,
- `alpha_rel` — suppression exponent,
- `chi_g` — RDCC visibility modification amplitude,
- `chi_Phi` — RDCC potential modification amplitude,
- `chi_rei` — RDCC reionization-bump amplitude.

These parameters mirror the analytical forms introduced in Appendix A.

B.3 Modifications to the Visibility Function

The standard visibility function is computed in `thermodynamics.c` as

$$g(z) = \frac{d\tau}{dz}\, e^{-\tau(z)}.$$

In RDCC, the visibility function is modified as

$$g_{\rm RDCC}(z) = g(z) \left[1 + \chi_g \left(\frac{k_{\rm rel}}{k_{\rm rei}(z)} \right)^{\alpha_g} \right], \tag{46}$$

where $k_{\rm rei}(z)$ is the characteristic reionization scale.

This modification is applied after the standard CLASS visibility calculation.

B.4 Modifications to the Optical Depth

The optical depth is computed in `thermodynamics.c` as

$$\tau(z) = \int dz' \, \frac{c\,\sigma_T\, n_e(z')}{(1+z')H(z')}.$$

In RDCC, the electron density is unchanged, but the large-scale potentials entering the ionization history are modified by the relaxation-suppressed growth rate.

A compact implementation is

$$\tau_{\rm RDCC}(z) = \tau(z)\left[1 + \chi_\tau \left(\frac{k_{\rm rel}}{k_{\rm rei}(z)}\right)^{\alpha_\tau}\right]. \tag{47}$$

This modification is applied before computing the visibility function.

B.5 Modifications to Large-Scale Potentials

The large-scale potential is computed in `perturbations.c` as

$$\Phi(k,z) \propto D(z)\,T(k).$$

In RDCC:

$$\Phi_{\rm RDCC}(k,z) = \Phi(k,z)\left[1 - \chi_\Phi \left(\frac{k}{k_{\rm rel}}\right)^{\alpha_\Phi}\right]. \tag{48}$$

This modification affects the quadrupole at reionization and therefore the reionization bump in polarization.

B.6 Reionization Bump in Polarization

The RDCC reionization bump is implemented in `perturbations.c` as

$$C_{\ell,\rm RDCC}^{EE,\rm rei} = C_\ell^{EE,\rm rei}\left[1 + \chi_{\rm rei}\left(\frac{\ell_{\rm rel}}{\ell}\right)^{\alpha_{\rm rei}}\right], \tag{49}$$

with $\ell_{\rm rel} = k_{\rm rel}\chi_{\rm rei}$.

This modification is applied after the standard CLASS polarization source integration.

B.7 Output and Post-Processing

The RDCC reionization outputs are written to:

- `cl_EE_rei.dat`,
- `visibility_rdcc.dat`,
- `tau_rdcc.dat`.

These files can be used for comparison with Planck, LiteBIRD, and future polarization surveys.

B.8 Summary

The CLASS implementation of RDCC reionization requires:

- RDCC-modified visibility and optical-depth kernels,
- RDCC-modified large-scale potentials,
- reionization-bump corrections in polarization,
- new CLASS parameters and data structures.

This implementation provides a complete numerical framework for computing RDCC reionization signatures and enables direct comparison with current and future CMB polarization experiments.

Appendix C: Numerical Settings and Validation Tests

This appendix summarizes the numerical settings, convergence tests, and validation procedures used to compute the RDCC reionization signatures presented in this Companion. The goal is to ensure reproducibility and to provide guidance for implementing RDCC reionization in CLASS and related analysis pipelines.

C.1 Numerical Resolution Requirements

Accurate computation of RDCC-modified visibility and optical-depth kernels requires sufficient resolution in redshift and conformal time.

Recommended settings:

- $\Delta z \leq 0.005$ for visibility-kernel integration,
- $\Delta\eta \leq 0.1\,\mathrm{Mpc}$ for polarization source functions,
- $k_{\max} \geq 5\,h\,\mathrm{Mpc}^{-1}$ for potential evolution,
- $\ell_{\max} \geq 200$ for the reionization bump in EE.

These settings ensure convergence of the RDCC-modified reionization spectra at the percent level.

C.2 Convergence Tests

We perform three classes of convergence tests:

(1) Redshift sampling. The visibility function $g(z)$ is sharply peaked. Convergence is achieved when:

$$\frac{\Delta C_\ell^{EE,\mathrm{rei}}}{C_\ell^{EE,\mathrm{rei}}} < 0.5\% \quad \text{for } \ell < 20.$$

(2) Potential evolution. The RDCC suppression factor

$$1 - \chi_\Phi \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha_\Phi}$$

requires fine sampling near $k \sim k_{\mathrm{rel}}$. Convergence is achieved when:

$$\frac{\Delta\Phi_{\mathrm{RDCC}}}{\Phi_{\mathrm{RDCC}}} < 0.3\%.$$

(3) Polarization source integration. The reionization bump is sensitive to the quadrupole at $z \sim 6$–10. Convergence is achieved when:

$$\frac{\Delta C_\ell^{EE}}{C_\ell^{EE}} < 0.2\% \quad \text{for } \ell < 30.$$

C.3 Validation Against ΛCDM

RDCC reionization must reduce to ΛCDM in the limit:

$$\chi_g, \chi_\Phi, \chi_{\rm rei} \to 0, \qquad k_{\rm rel} \to \infty.$$

We validate:

- $g_{\rm RDCC}(z) \to g(z)$ at the $< 0.1\%$ level,
- $\tau_{\rm RDCC}(z) \to \tau(z)$ at the $< 0.1\%$ level,
- $C_{\ell,\rm RDCC}^{EE,\rm rei} \to C_\ell^{EE,\rm rei}$ at the $< 0.2\%$ level,
- the quadrupole evolution matches CLASS ΛCDM at the $< 0.1\%$ level.

These checks ensure that RDCC is a controlled extension of ΛCDM.

C.4 Validation of RDCC Reionization Signatures

We validate the characteristic RDCC signatures:

(1) Enhanced large-scale polarization. For $\ell < \ell_{\rm rel}$:

$$C_{\ell,\rm RDCC}^{EE,\rm rei} > C_\ell^{EE,\rm rei} \quad \text{by } 10\% - -25\%.$$

(2) Smoother reionization bump. The RDCC bump is broader and less sharply peaked.

(3) Reduced small-scale reionization fluctuations. For $\ell > \ell_{\rm rel}$:

$$C_{\ell,\rm RDCC}^{EE,\rm rei} < C_\ell^{EE,\rm rei} \quad \text{by } 5\% - -15\%.$$

(4) Visibility-kernel coherence. The RDCC visibility kernel satisfies:

$$g_{\rm RDCC}(z) > g(z) \quad \text{for } z \sim 6\text{–}10.$$

C.5 Numerical Stability and Error Control

To ensure numerical stability:

- enforce positivity of $g_{\rm RDCC}(z)$ and $\tau_{\rm RDCC}(z)$,
- apply spline smoothing to RDCC suppression factors,
- ensure smooth transitions in the quadrupole evolution,
- use adaptive redshift sampling near the visibility peak.

These steps prevent numerical artifacts in the RDCC reionization spectra.

C.6 Summary

This appendix provides:

- recommended numerical settings for RDCC reionization,
- convergence tests for redshift sampling and potential evolution,
- validation against ΛCDM,
- verification of RDCC reionization signatures,
- guidelines for stable numerical implementation.

These procedures ensure that RDCC reionization predictions are accurate, reproducible, and consistent with the analytical and numerical framework developed in this Companion.